

Simon Yan

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EDUCATION

University of California Santa Barbara

Goleta, California

B.S. in Computer Science

Expected Graduation, Dec 2026

- o GPA: 3.70/4.00, Dean's Honors, Engineering Honors Program
- o Related Coursework: Data Structures & Algorithms, Object-Oriented Design, Machine Learning & Deep Learning, Linear Algebra, Probability & Statistics, Entrepreneurship

EXPERIENCE

University of California Santa Cruz

Santa Cruz, California

Machine Learning Research Intern

Dec 2021 – May 2022

- Researched electric load forecasting and explored impact of renewable energy sources on power grid performance
- Optimized Python scripts using polynomial regression models and ARIMA to achieve over 95% accuracy
- Delivered analysis of training methods and model performance, provided insight on improving forecasting accuracy

Anti-Tremor Glove

Goleta, California

Project Lead & Software Team

Nov 2024 – Present

- Coordinated diverse team of students to conceptualize and market a glove that alleviates Parkinsonian symptoms
- Interviewed patients, clinicians, and insurance providers, to gain market validation and improve product-market fit
- Combined flex sensors for hand curvature and pressure measurements with reinforcement learning to provide adaptive grip enhancement and stability

PROJECT EXPERIENCE

Speech & Debate Bot

Python, TensorFlow, OpenCV, React.js

Nov 2024 – Present

- Leveraged speech-to-text APIs and NLP models to assess speech argument strength and language usage
- Modeled sequence-based neural network to assign scores at each timestep based on speech and gesture quality
- Collected training and testing dataset by dividing online speech and debate videos into short clips
- Built a web application for users to record or upload speech videos and receive personal score and feedback

Worldly Bites

Python, Flask API, TensorFlow, React.js, JavaScript, Expo Go

Mar 2024 – Aug 2024

- Developed cultural discovery mobile app connecting users with diverse recipes from around the world
- Enhanced user experience by integrating both manual input and camera scanning that recognizes over 40 ingredients
- Designed TF-IDF model to recommend top 5 recipes out of a database of 13,000 to increase user engagement through personalized suggestions
- Coded a cuisine prediction feature that classifies over 20 cuisines based on ingredients to promote cultural diversity

Red Wine Quality Calculator

Python, React.js, JavaScript, SingleStore

Jan 2024 – Feb 2024

- Created a web app with interactive sliders and graphs allowing users to adjust key features influencing wine quality
- Analyzed red wine quality dataset and identified linear regression as the best model, reaching over 97% accuracy
- Streamlined user winemaking experience by highlighting four key variables most impacting wine quality out of eleven
- Achieve second place in SingleStore x Data Science UCSB Datathon, showcasing innovation and teamwork

Brainwave

Firebase, Firestore Cloud, Flutter, Dart

Jan 2024

- Developed a social media style mobile app for students to easily find and join study groups
- Added a Firebase backend to securely manage user data and facilitate real-time chats with Firestore Cloud
- Incorporated Google Authentication for secure user registration and login to ensure app security and user safety
- Designed interactive features like post liking and private chats to foster collaboration and community among students

SKILLS

Programming: C++, Python, JavaScript, HTML/CSS, React.js, TensorFlow, PyTorch, Git, Visual Studio Code, Dart